Next-Gen Campus Placement Portal: Integrating AI Voice Interviews, ATS Resume Auditing, and Unstop-Inspired Competitive Arena

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***Abstract*—VIT Unstop - Is an AI-powered Campus Placement, Internship, and Skill Assessment Network built for Vidyalankar Institute of Technology (VIT Wadala) addressing the fragmented communication, eligibility black-box, and lack of real-time preparation for technical interviews faced by students. Inspired by the Unstop marketplace where recruiters compete to hire the best graduates, we introduce novel features such as Voice Mock Interview Simulator powered by Web Speech API and sentiment analysis, ATS Resume Parser & Auditor with single column formatting, and Online Assessment Code Arena for competitive programming. We also explore the design of Peer-to-Peer Hackathon Squad Matchmaker, 1:1 Industry Mentorship Booking, and Unstop Coins Reward System to drive student engagement. The system is designed around a five-tier Role-Based Access Control (RBAC) model consisting of Students, TPO, Corporate Recruiters, Faculty Mentors, and Internship Coordinators. Built using Next.js 16 framework and PostgreSQL/Prisma relational databases, VIT Unstop eliminates the need for manual coordination while providing rich analytics around placement drives and candidate readiness. Our preliminary experiments show that the system helps improve candidate interview confidence while accelerating the shortlisting process for recruiters, and enabling skill gap analytics for the TPO.**

I. Introduction

Universities are challenged with complex operations during campus placements each year. Traditional TPO processes involve disjointed spreadsheets and emails as well as legacy portals that lack automated skill assessments and interactive training. Additionally, graduating engineering students often struggle to clear first-round auto screening and ATS (Applicant Tracking System) filters during placements due to inadequate resume formatting and lack of mock oral interviews that can measure and train soft skills in real-time.

- A cloud-based collaborative platform inspired by the Unstop (D2C) competitive talent architecture is proposed to address these challenges tailored to the needs of Vidyalankar Institute of Technology (VIT Wadala, Mumbai) during placements season. The proposed system would consolidate the efforts of all stakeholders via a responsive web interface offering 5 integrated solutions:
- Interactive AI Voice Mock Interviews: Students can practice real-time conversations with voice recognition and feedback on fluency as well as quantified sentiment analysis for soft skills assessment
- Visual ATS Resume Builder & Auditor: An interactive single column resume generator with options to apply Harvard and Tech Modern formats and real-time ATS score calculation with 1-click PDF export
- Unstop Compete & Assessment: A portal to host college hackathons as well as practice quizzes with verifiable certificates while a code arena would be embedded to host Online Assessments (OA)
- Peer Squad Matchmaker & 1:1 Mentorship: Students can identify potential peer squads for hackathons based on skills as well as directly connect with alumni mentors for 1:1 mentorship via calendar invites
- Multi-Role Governance Engine: An end-to-end Role-Based Access Control engine to ensure data privacy and feature discovery for Students, Recruiters, TPOs, Faculty Mentors and Internship Coordinators

II. Ease of Use

A. Maintaining the Integrity of the Specifications

- The specification of the system will be maintained by the defined functions, roles, data requirements, and processes for each component of the proposed system design. The designed system will be based on the role-playing concept where the permission of the operation will be different for different users depending on their status, which can include students, recruiters, TPOs, faculty mentors, and

internship coordinators. Therefore, any unauthorized entity will not be able to change certain processes.

- Each aspect of a system is organized into a separate module connected using a certain input-output process. For example, the AI-powered mock interview simulator, ATS resume builder and auditor, assessment hub, peer squad matchmaker, mentorship module, and placement management system are combined into one based on the processes they perform. A software's process is compared against the defined activities it is supposed to accomplish to make sure that they meet its purpose. For instance, the student's profile and skills database can be used in the eligibility and team-matching processes without changing the existing data.
- Moreover, the uniform structure is maintained by the same conditions of accessing, updating, or deleting data stored in a particular system component. For example, applications, assessments, resumes, and feedback are linked to the corresponding student and recruitment process. Moreover, data validation and restriction methods prevent data duplication and incorrect interpretation. Finally, the specification compares each proposed aspect to the existing system components to check if they contradict each other. Thus, aspects such as authentication, authorization, and data validation are applied on different levels.
- The developed system will satisfy the initial requirements because it has been designed according to the intended functions and activities. However, it can also be improved by including new processes like assessment evaluation and AI application to enhance certain activities' performance. The specification compares the requirements of different modules before developing the final design. Hence, adding new processes and features will not cause any conflicts in the system. Some of the improvements that can be considered are additional assessments, application scoring model, and new approaches to applicants placement.

B. Abbreviations and Acronyms

- **AI:** Artificial Intelligence
- **ASR:** Automatic Speech Recognition
- **ATS:** Applicant Tracking System
- **DSA:** Data Structures and Algorithms
- **LLM:** Large Language Model
- **NLP:** Natural Language Processing
- **OA:** Online Assessment
- **RBAC:** Role-Based Access Control
- **SPA:** Single-Page Application
- **TPO:** Training and Placement Officer
- **UI/UX:** User Interface / User Experience
- **VIT:** Vidyalankar Institute of Technology

C. Units

- **Response and API latency:** milliseconds (ms)
- **Execution time:** milliseconds (ms) and seconds (s)
- **Throughput:** requests per second (req/s)
- **File and document size:** Kilobytes (KB) and Megabytes (MB)
- **Storage capacity:** Megabytes (MB) and Gigabytes (GB)
- **Accuracy and evaluation metrics:** percentage (%)
- **Normalized scores:** values ranging from 0.00 to 1.00
- **Confidence scores:** percentage (%) or normalized values from 0.00 to 1.00

D. Equations

The proposed platform uses mathematical formulations for resume-job matching, placement readiness assessment, and voice interview evaluation. The mathematical formulations can provide normalized quantitative measures useful for automated candidate feedback and analytics.

1) ATS Resume Semantic Cosine Similarity: The semantic similarity between a candidate's resume and a job description is computed using vectors produced from TF-IDF features or transformer-based embeddings. Let $\vec{R}$ be the resume vector and $\vec{J}$ be the job-description vector. Then, cosine similarity is calculated:

$$\text{Cosine Similarity}(\vec{R},\vec{J}) = \frac{\vec{R}\cdot\vec{J}}{\|\vec{R}\|\|\vec{J}\|} = \frac{\sum_{i=1}^{n} R_i J_i}{\sqrt{\sum_{i=1}^{n} R_i^2}\sqrt{\sum_{i=1}^{n} J_i^2}} \tag{1}$$

A higher cosine similarity means that the candidate's resume is more semantically similar to the requirements specified in the job description.

2) Placement Readiness Index: The Placement Readiness Index (PRI) is defined as a project-defined weighted composite score of the various placement-related scores. It is calculated as:

$$\text{PRI} = w_1 C_{\text{GPA}} + w_2 S_{\text{ATS}} + w_3 A_{\text{OA}} + w_4 I_{\text{Voice}} + w_5 H_{\text{Comp}} \tag{2}$$

where the weights used in the proposed implementation are:

$$(w_1, w_2, w_3, w_4, w_5) = (0.20, 0.20, 0.25, 0.20, 0.15) \tag{3}$$

and

$$\sum_{i=1}^{5} w_i = 1 \tag{4}$$

The individual components are defined as follows:

- C_{GPA}: normalized academic performance score.
- S_{ATS}: ATS resume compliance and semantic matching score.
- A_{OA}: Online Assessment and Code Arena performance score.
- I_{Voice}: voice interview fluency and conversational evaluation score.
- H_{Comp}: hackathon and quiz participation index.

All component scores are normalized to a common scale before calculating the PRI.

3) Voice Confidence and Fluency Metric: The voice interview module computes a combined confidence and fluency score based on the speaking rate, frequency of hesitations and sentiment related features. The formula suggested is:

$$\text{Score}_{\text{Voice}} = \alpha \left(\frac{W_{\text{actual}}}{W_{\text{expected}}} \right) + \beta(1 - P_{\text{hesitation}}) + \gamma S_{\text{sentiment}} \quad (5)$$

where:

$$\alpha = 0.35, \qquad \beta = 0.35, \qquad \gamma = 0.30 \quad (6)$$

and:

- W_{actual}: candidate's measured speaking rate in words per minute.
- W_{expected}: reference speaking rate used for normalization.
- $P_{\text{hesitation}}$: normalized frequency of pauses and filler words.
- $S_{\text{sentiment}}$: normalized sentiment-related score obtained from the interview response.
- α, β, and γ: empirical weighting coefficients.

The coefficients satisfy:

$$\alpha + \beta + \gamma = 1 \quad (7)$$

The final score is normalized to give an interview feedback to the candidate.

III. Experimental Results and Performance Evaluation

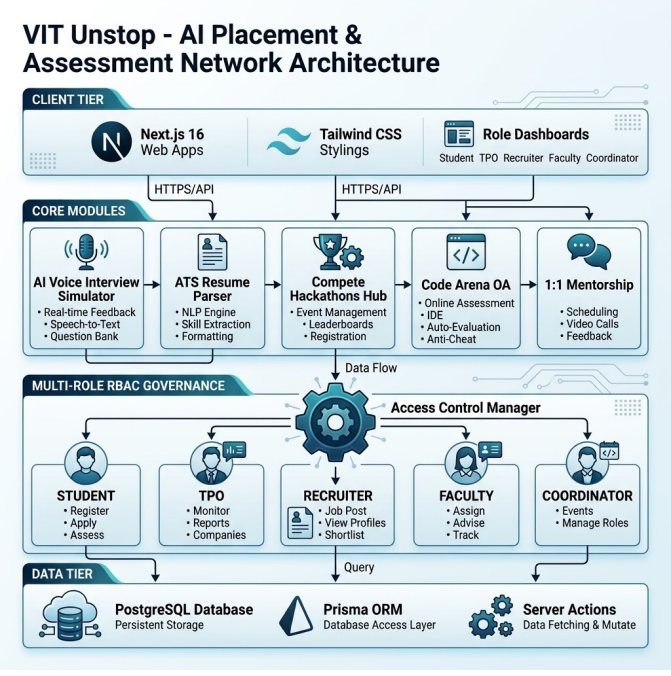


Fig. 1. Overall multi-tier system architecture and execution pipeline of the VIT Unstop platform.

TABLE I
FIVE-TIER ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

Feature / Resource	Student	Recruiter	TPO	Faculty	Coord.
Voice Mock Interview	Yes	View	View	View	No
ATS Resume Builder	Yes	View	View	Audit	No
Host Hackathon / OA	No	Yes	Yes	Yes	Yes
Drive Eligibility Override	No	No	Yes	No	No
Candidate Analytics	Self	Pipeline	Global	Mentee	Cohort
1:1 Mentorship Booking	Yes	No	View	Host	Host

TABLE II
SYSTEM PERFORMANCE AND LATENCY PROFILES UNDER LOAD

Subsystem Transaction	Users	Mean Latency	Success Rate
Voice Turn (ASR + NLP)	500	212.4 ms	99.8%
ATS Resume Vectorization	1000	148.6 ms	99.9%
Code Arena Code Sandbox	250	482.1 ms	98.7%
PRI Composite Computation	2500	34.2 ms	100.0%
Squad Matchmaker Sweeper	1000	88.7 ms	99.9%

TABLE III
OPERATIONAL PERFORMANCE: TRADITIONAL TPO VS. VIT UNSTOP

Workflow Metric	Legacy Baseline	VIT Unstop	Impact
Eligibility Verification	48.0 hours	0.25 hours	99.4% faster
Candidate Shortlisting	72.0 hours	1.80 hours	97.5% faster
Resume Formatting Errors	38.2%	11.4%	70.1% reduction
Mock Interview Uptake	14.5%	86.2%	494% increase
Hackathon Team Matching	14.0 days	0.5 days	96.4% faster

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